The set of true sentences of arithmetic is not definable in arithmetic

The Open Logic Project contributors

openlogic:incompleteness/incompleteness-provability:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel thm:tarski of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the fixed-point (diagonal) lemma for Q
- the Goedel numbering gn of sentences
- definability of a set of numbers by a formula in the standard model

Theorem 0.1. *The set of true !!sentences of arithmetic is not definable in arithmetic.*

Proof. Suppose $!D(x)$ defined it, i.e., $N \models !A$ iff $N \models !D(\mathrm{gn}\,!A)$. By the fixed-point lemma, there is a formula $!A$ such that $\mathrm{Th}Q \vdash !A \leftrightarrow \neg !D(\mathrm{gn}\,!A)$, and hence $N \models !A \leftrightarrow \neg !D(\mathrm{gn}\,!A)$. But then $N \models !A$ if and only if $N \models \neg !D(\mathrm{gn}\,!A)$, which contradicts the fact that $!D(y)$ is supposed to define the set of true statements of arithmetic. □

Proof. Suppose $!D(x)$ defined it, i.e., $N \models !A$ iff $N \models !D(\mathrm{gn}\,!A)$. By the fixed-point lemma, there is a formula $!A$ such that $\mathrm{Th}Q \vdash !A \leftrightarrow \neg !D(\mathrm{gn}\,!A)$, and hence $N \models !A \leftrightarrow \neg !D(\mathrm{gn}\,!A)$. But then $N \models !A$ if and only if $N \models \neg !D(\mathrm{gn}\,!A)$, which contradicts the fact that $!D(y)$ is supposed to define the set of true statements of arithmetic. □